

CO-2 ASSESSMENT-TOOL-1

Scenario-Based Industrial Critical Questions

COURSECODE:CSA-0924-JAVA

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1. A shopping-cart service serialises collection data to JSON. The order of map entries unexpectedly changes between runs and breaks a snapshot test. Explain why relying on HashMap iteration order is unsafe and propose an order-preserving alternative. [Map Ordering]

Answer:

AIM:

To demonstrate why **HashMap iteration order is not guaranteed** and how **LinkedHashMap** preserves insertion order, making it suitable for JSON serialization and snapshot testing.

Pseudo Code:

START

Create a HashMap

Insert product entries

Display HashMap entries

(Note: Order may change in different runs)

Create a LinkedHashMap

Insert the same product entries

Display LinkedHashMap entries

(Order remains same as insertion order

END.

Code:

```
import java.util.HashMap;
```

```
import java.util.LinkedHashMap;
```

```
import java.util.Map;
```

```
public class MapOrderingDemo {
```

```
    public static void main(String[] args) {
```

```
        // HashMap (Order not guaranteed)
```

```
        HashMap<String, Integer> hashMap = new HashMap<>();
```

```
        hashMap.put("Laptop", 1);
```

```
        hashMap.put("Mouse", 2);
```

```
        hashMap.put("Keyboard", 1);
```

```
        hashMap.put("Monitor", 1);
```

```
        System.out.println("HashMap Output:");
```

```
        for (Map.Entry<String, Integer> entry : hashMap.entrySet()) {
```

```
        System.out.println(entry.getKey() + " = " + entry.getValue());
    }

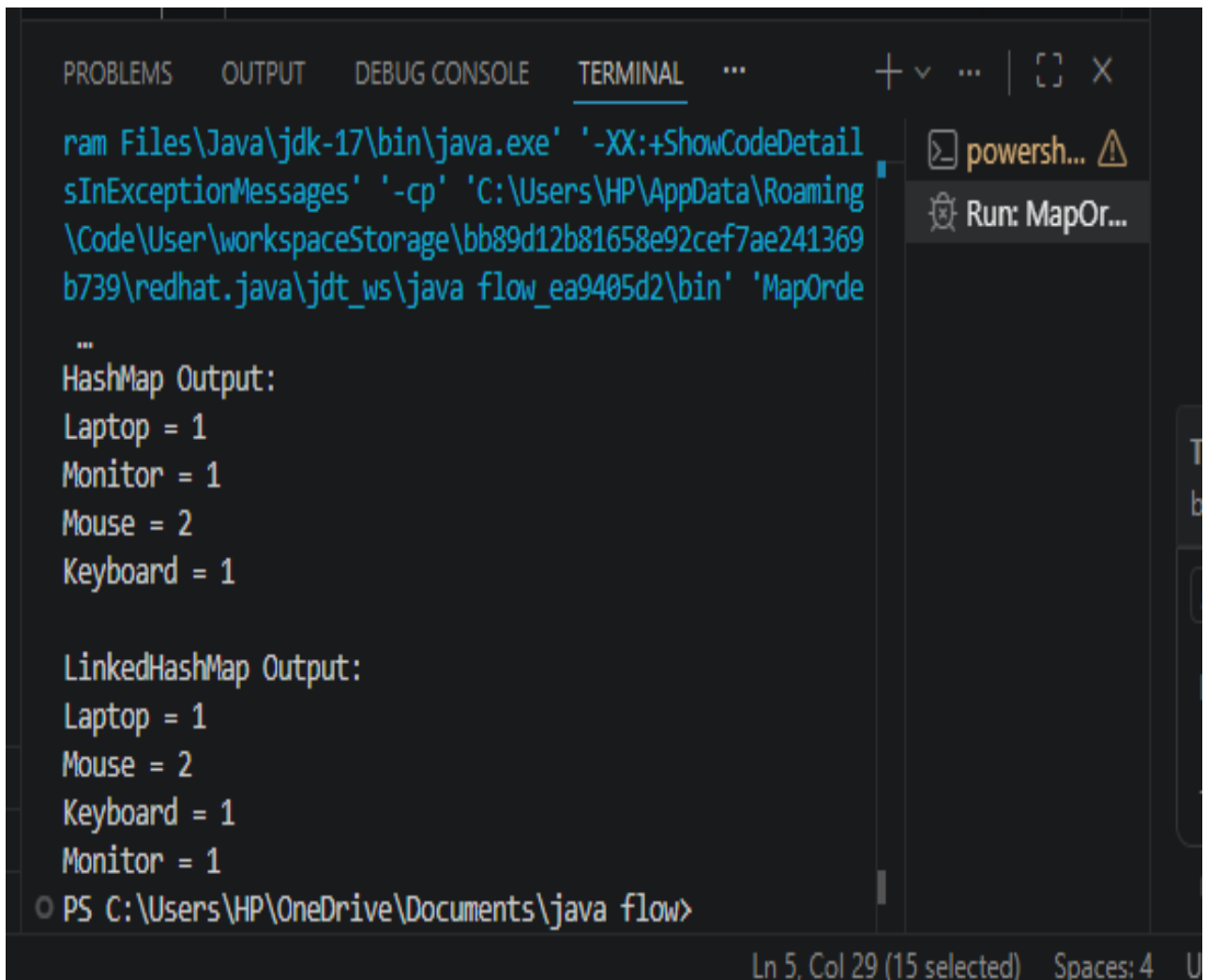
    System.out.println();

    // LinkedHashMap (Insertion order preserved)
    LinkedHashMap<String, Integer> linkedMap = new
    LinkedHashMap<>();

    linkedMap.put("Laptop", 1);
    linkedMap.put("Mouse", 2);
    linkedMap.put("Keyboard", 1);
    linkedMap.put("Monitor", 1);

    System.out.println("LinkedHashMap Output:");
    for (Map.Entry<String, Integer> entry : linkedMap.entrySet()) {
        System.out.println(entry.getKey() + " = " + entry.getValue());
    }
}
}
```

Output:



The screenshot shows an IDE terminal window with the following content:

```
ram Files\Java\jdk-17\bin\java.exe' '-XX:+ShowCodeDetail
sInExceptionMessages' '-cp' 'C:\Users\HP\AppData\Roaming
\Code\User\workspaceStorage\bb89d12b81658e92cef7ae241369
b739\redhat.java\jdt_ws\java flow_ea9405d2\bin' 'MapOrde
...
HashMap Output:
Laptop = 1
Monitor = 1
Mouse = 2
Keyboard = 1

LinkedHashMap Output:
Laptop = 1
Mouse = 2
Keyboard = 1
Monitor = 1
PS C:\Users\HP\OneDrive\Documents\java flow>
```

On the right side of the terminal, there is a panel with the following items:

- A button labeled "powersh..." with a warning icon.
- A button labeled "Run: MapOr..." with a play icon.

The status bar at the bottom right indicates "Ln 5, Col 29 (15 selected) Spaces: 4 U".